

KANISHKA R. JHA

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EDUCATION

SIES GST B.E in <i>Electronics & Computer Engineering</i> Arun Muchhala Polytechnic Diploma in <i>Computer Engineering</i> .	Aug 2022 – Jun 2025 (CGPA: 7.4) Aug 2019 – Jul 2022 (Final year: 82%)
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SKILLS

- **Certifications (In Progress: CRTP)**
- Active Directory Security Assessment
- Windows Authentication & Identity Security (NTLM / Kerberos)
- Privilege Escalation Assessment (Windows & Linux)
- Internal Network Penetration Testing
- Lateral Movement & Access Propagation Analysis
- Enterprise Network Segmentation & Pivoting Assessment
- Web Application Security Assessment (OWASP Top 10)
- API Security Testing (REST & GraphQL)
- SAST using SonarQube
- Application Security Testing Methodology
- Threat Modeling & Risk Validation (MITRE ATT&CK / STRIDE)
- Llama Guard (AI Guardrails integration)
- Owasp top 10 for LLM Security Testing (Prompt Injection & Adversarial Testing)

EXPERIENCE

Security Engineer, Christelijke Mutualiteit *Jan 2026 - Present*

- Performed penetration testing on **web applications and AWS** environments to identify vulnerabilities and misconfigurations.
- Collaborated with the red team on AppSec assessments, identifying issues in authentication and input validation.
- **Creating / Designing** vulnerable / misconfigured scenario-based labs.
- Conduct internal security audit for **Active Directory**.
- Created detection use cases for **Active Directory** based attacks.
- **Authored technical blogs** on vulnerabilities, exploitation techniques, and remediation strategies.

Penetration Tester, Cognizant Technology Solutions *Aug 2025 - Jan 2026*

- Performed internal network and web application penetration testing across Windows-based enterprise environments.
- Conducted Active Directory enumeration to identify security weaknesses and potential attack paths.
- Identified and reported **25+ vulnerabilities** across internal systems, including issues in authentication, access control, and misconfigurations.
- Assessed privilege escalation opportunities and lateral movement paths to evaluate overall security posture.
- Tested internal web applications for vulnerabilities related to input validation, authentication flaws, and improper access controls

Cybersecurity Engineer - Technoxi, Texas *May 2025 - Aug 2025*

- Conducted security assessments simulating attacker behavior from application compromise to internal system impact.
- Worked on a **project-based engagement**, collaborating with the **HP security team** to deliver security solutions and assessments
- Contributed to the delivery of **enterprise security implementations**
- Identified **23+ high-risk vulnerabilities** and supported remediation to strengthen overall security posture
- Demonstrated real-world attack impact and validated security control effectiveness.

AI-Security Intern - Jio, Mumbai *Jun 2024 - Oct 2024*

- Designed **LLM security test suites** based on **OWASP Top 10 for LLMs**, simulating real-world adversarial attacks.
- Performed **stress and adversarial testing** using **Apache JMeter** and **Locust**.
- Identified vulnerabilities like **prompt injection and data leakage**, improving model safety.
- Contributed to **Llama Guard integration** for enforcing AI safety guardrails.

Security Intern, Technoxi, Texas *Feb 2024 - May 2024*

- Deployed advanced security measures, reducing incidents by **14%**.
- Built AI chatbot for Cisco ISE & Umbrella APIs, cutting response time by **33%**.
- Automated data fetching & monitoring, saving **44% manual effort**.
- Conducted web app pentests, mitigating **147+ vulnerabilities**.
- Integrated Umbrella & ISE APIs with Cisco, improving detection by **27%**.
- Delivered tailored security solutions to global clients, ensuring **96% satisfaction**.
- Supported security compliance efforts, contributing to **ISO 27001 readiness**.

Infrastructure Security Intern, BSES Reliance ADA GROUP, Delhi *Jun 2023 - Aug 2023*

- Triaged internal vulnerability reports and reduced false positives by **17%** through validation and reproduction testing.
- Identified and documented **46+ vulnerabilities** across multiple web applications and APIs.
- Performed web application and API security testing focusing on authentication, authorization, and input validation flaws.
- Conducted source code review to detect logic errors, insecure data handling, and improper validation.
- Collaborated with development teams to remediate issues and explained root cause and secure implementation practices.
- Verified fixes after patching and confirmed vulnerabilities could not be re-exploited in production environments.
- Recognized recurring vulnerability patterns and recommended secure coding improvements to prevent similar issues.

TOOLS & SOFTWARE PROJECTS

1. Custom Service & Port Enumeration Scanner

- Developed a multi-threaded service discovery and enumeration tool to identify exposed network services and application entry points.
- Implemented TCP connect and banner-grabbing mechanisms to detect running services and their versions.
- Added service fingerprinting to recognize web servers, database services, remote access protocols, and misconfigured internal services.
- Designed targeted checks for common insecure configurations such as publicly accessible management interfaces and exposed development services.
- Integrated HTTP probing to automatically discover web applications and enumerate accessible endpoints.
- Used the tool during lab assessments to map attack surfaces and prioritize targets for further application and authentication testing.
- Demonstrated how unnecessary exposed services can increase attack surface and enable initial foothold within internal environments..

2. Vulnerable Streaming Platform Lab

- Built a deliberately vulnerable web application simulating a media streaming platform to study real-world attack scenarios.
- Implemented user authentication, subscription management, content access controls, and administrative panels to mimic production systems.
- Intentionally introduced common security weaknesses including broken access control, insecure session handling, and improper input validation.
- Used the lab to practice exploitation and understand how application vulnerabilities could expose user data and administrative functionality.
- Simulated attacker workflows from account compromise to unauthorized content access and privilege misuse.
- Tested remediation approaches by patching vulnerabilities and validating fixes after mitigation.
- Used the platform to study how an application compromise could potentially affect backend services and stored sensitive information.

3. EKA: Trouble29 - Full-Surface Security Automation | Python, Recon, Web & Network Testing, AWS

- Modular toolkit for end-to-end assessments: discovery → validation → reporting.
- Discovery & attacks: subdomain/vhost enumeration, OSINT/URL harvesting, port/service scanning, directory brute-force, fuzzing, payload injection, web-cache deception, SSRF, open redirects, auth flaws, and more.
- Hosted on AWS with VPC, public/private subnets, NAT Gateway, ALB, and Auto Scaling; enforced least-privilege access using IAM roles and Security Groups
- Produces triage-ready reports with sandboxed execution, rate limits, and false-positive reduction.

ACHIEVEMENTS

1. Industry Vulnerability Disclosure

- Achieved **Hall of Fame recognition** from **Philips**: Discovered a vulnerability exposing PII of 3 lakh users.
- Achieved **Hall of Fame recognition** from **NASA**: Reporting sensitive data exposure through an insecure endpoint.
- Achieved **Hall of Fame recognition** from **Oracle**: Identified vulnerabilities providing access to internal files and databases.
- Achieved **Hall of Fame recognition** from **Latam Airways**: Uncovered critical vulnerabilities.
- Received a **Letter of Recognition** from **NASA**: Uncovered Proprietary Source Code of Internal Software's.

2. Security Research & Conferences

- Published a research paper on the EKA tool, which was selected for presentation at the IEEE International Conference on Advanced Networks and Telecommunications Systems (**ASIANCON**).
- Selected in **@NULLCON** International Conference in Goa for outstanding contributions to cybersecurity.

3. Competitions & Technical Activities

- Won **1st place @ Smart India Hackathon**, representing the college at the national level @IIT Jammu.
- Secured **2nd position** at DomeCTF, by c0c0n.
- Secured **4th position** at the M.H. Saboo Siddik College Hackathon (ERR_404 still not found).
- Secured selection in the distinguished Govt Kavach Hackathon, representing the college with innovative solutions.
- Recognized as a Swag recipient from HacktoberFest for contributing 4 pull requests to eligible repositories.
- Secured **14th position** in the Downunder DUCTF 4.0 challenge. Achieved first place at the Smart India Hackathon, representing the college at the national level at IIT Jammu.
- Secured second position at DomeCTF, hosted by c0c0n.
- Attained fourth position at the M.H. Saboo Siddik College Hackathon (ERR_404 still not found).
- Earned selection in the distinguished Government Kavach Hackathon, representing the college with innovative solutions.
- Recognized as a Swag recipient from HacktoberFest for contributing four pull requests to eligible repositories.
- Secured fourteenth position in the Downunder DUCTF 4.0 challenge.